

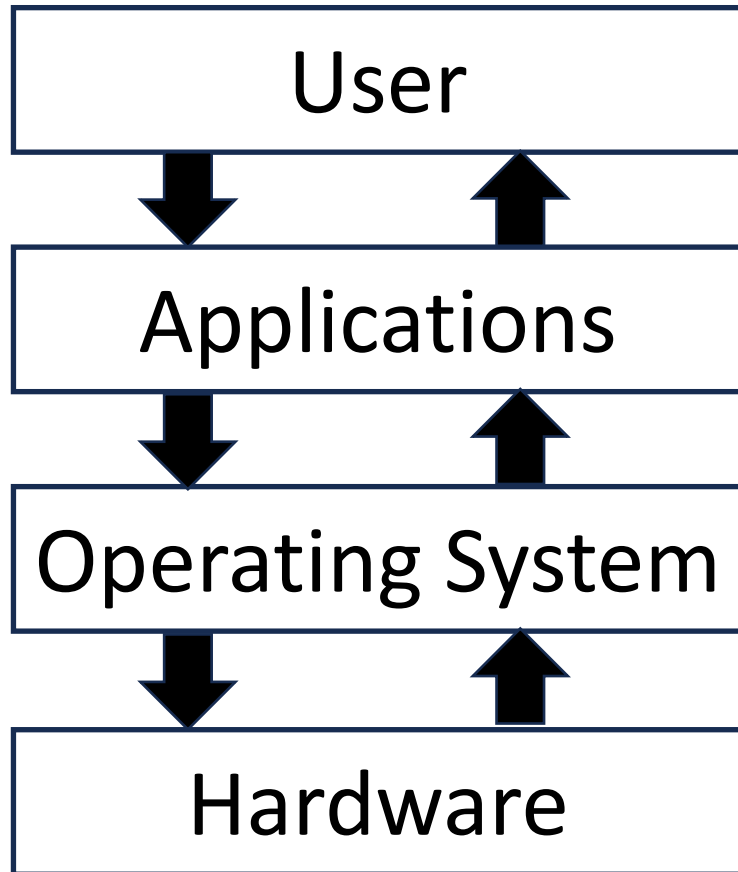
Real-Time Operating System (RTOS)

What is operating system?

It is a software layer between hardware and the application, which:

- Has direct access to hardware.
- Manage the hardware according to predefined rules & policies.
- Hide hardware complexity from the application software.
- Offers multitasking / Resource & data management.

What is an operating system?



Microsoft office, VLC, GCC, C/C++/Python prog

Windows / Linux

PC Motherboard

Examples of Operating Systems

- Microsoft windows
- Linux
- MAC OS
- Android
- iOS
- RTX
- FreeRTOS ...

What is RTOS?

- RTOS is multitasking operating system designed specifically to run applications with very **precise timing & high degree of reliability.**
- Or An RTOS is multitasking operating system designed to manage hardware resources and execute tasks within a guaranteed time frame.

What is RTOS?

RTOS = Real-Time + Operating System

What is Real-time System?

- Real-time systems are those systems which responds to external events in a Timely fashion.
- Types of Real-time systems:
 - Hard real-time system
 - Soft real-time system

Hard real-time system

- A system that meets the deadline all the times **strictly**, even under a worst case system load.
- Missing a deadline even a single time may result in fatal injury or loss of human life.
- Eg: Airbag system, Missile Approach Warning in Military Aircrafts, Autopilot in CARs & Aeroplanes, Missile...

Soft real-time system

- Missing a deadline does not result in fatal injury.
- But system should process the input & generate the result.
- Eg: Live video streaming in CCTV where missing few frames are negligible, Washing machine, Video games ...

Hard real-time vs Soft real-time system

No	Hard Real-Time System	Soft Real-Time System
1.	Missing a deadline leads to catastrophic failure.	Meeting deadlines is preferred but not always mandatory.
2.	Deterministic behaviour is mandatory.	Performance may degrade if deadlines are missed.
3.	Used in Life safety systems	Used In general systems
4.	Eg: Flight control & autopilot in Avionics, Pacemaker & Ventilators in Medical devices...	Eg: Video streaming, ATM machines, Weather forecasting...

Why is an RTOS designed?

- An RTOS is designed for time-sensitive systems, where tasks must be completed within specific time constraints.

Goal/purpose of designing RTOS

- An RTOS's main goal is to ensure that time-sensitive tasks meet strict timing deadlines, preventing delays that could lead to system failure or malfunction.

RTOSs are commonly used in applications like:

- Automotive systems
- Robotics
- Medical devices
- Transportations
- Aerospace & defence
- Consumer electronics...

Examples of time sensitive embedded systems

1. Medical Devices: Devices like pacemakers, ventilators, and infusion pumps require precise timing to monitor and deliver life-saving treatments in real time.

2. Automotive Systems: In modern vehicles, systems like anti-lock braking systems (ABS), airbags, and electronic stability control (ESC) rely on RTOS to respond quickly to changing conditions.

3. Robotics: Industrial robots and drones use RTOS to ensure precise movements and coordination in real-time tasks.

4. Aerospace and Defense: Systems like flight control systems, missile guidance, and satellite communication require real-time processing to ensure safety and mission success.

5. Consumer Electronics: Devices such as smart cameras, gaming consoles, and smart speakers rely on RTOS to ensure low-latency performance and real-time user interaction.

6. Transportation Systems: Systems in trains, traffic lights, and air traffic control use RTOS to ensure safe and timely operations.

Key features of RTOS

1. Deterministic behaviour
2. High performance
3. Safety and security
4. Priority based scheduling
5. Small footprint

Determinism: Repeating an input will result in the same output.

High performance: RTOS systems are fast and responsive, often executing actions within a small fraction of the time needed by a general OS.

Safety and security: RTOSes are frequently used in critical systems when failures can have catastrophic consequences, such as robotics or flight controllers. To protect those around them, they must have higher security standards and more reliable safety features.

Priority-based scheduling:

Priority scheduling means that actions assigned a high priority are executed first, and those with lower priority come after. This means that an RTOS will always execute the most important task.

Small footprint: Versus their hefty general OS counterparts, RTOSes weigh in at just a fraction of the size. For example, Windows 10, with post-install updates, takes up approximately 20 GB. VxWorks®, on the other hand, is approximately 20,000 times smaller, measured in the low single-digit megabytes.

Examples of RTOS

Open source

- FreeRTOS
- RTX / RTX-Tiny
- uCOS
- RTLinux
- Zephyr...

Commercial

- QNX Neutrino
- ThreadX
- VxWorks
- Integrity...

RTOS vs GPOS

Real-Time OS	General Purpose OS
1. Deterministic OS	1. Non deterministic OS
2. Bare-metal like OS	2. Generally includes user friendly GUI
3. Task scheduling is always priority based	3. Task scheduling is not priority based always
4. Specially designed for Real-time embedded systems	4. Designed for PCs & Servers

Real-Time OS	General Purpose OS
5. Small size OS	5. Comparatively larger in size
6. No priority inversion	6. Priority inversion
7. Kernel used here is micro-kernel	7. Kernel used here is monolithic or hybrid.
8. Always generates faster & quick responses for internal & external events	8. Some cases non real-time response
9. Eg: RTX, QNX, FreeRTOS, Zypher, ThreadX...	9. Eg: Windows 7/10/11, Ubuntu,MAC OS, iOS, Android

Types of Kernels

1. Monolithic Kernel (Linux & Unix)
2. Micro Kernel (RTOS)
3. Hybrid Kernel (Windows, MAC OS & iOS)

